

purdue-l1-basic-controllers brief

Purdue Basic Controllers (L1) — Cross-Vendor Threat & Detection Brief

Scope: Purdue Level 1 — the *Basic Control* layer — comprises the programmable logic controllers (PLCs), remote terminal units (RTUs), intelligent electronic devices (IEDs), protective relays, variable-frequency drives (VFDs), and distributed I/O modules that physically read sensors and command actuators in real time. This brief cross-cuts Rockwell Automation (ControlLogix / CompactLogix / Micro800 / PowerFlex), Siemens (SIMATIC S7-300/400/1200/1500, SINAMICS, SIPROTEC, ET200), Schneider Electric (Modicon Quantum / M340 / M580 / M221, Altivar, Sepam/MiCOM), Honeywell (C300, ControlEdge RTU), SEL (RTAC, 300/400/700-series relays), and ABB (REF/REL/RET). L1 sits between L0 (field instrumentation — sensors, actuators, valves) and L2 (HMI / Supervisory). It is the *last programmable layer before physics*: a compromised L1 device can damage equipment, injure people, or falsify the view sent upward to L2/L3. L1 also frequently shares Ethernet segments with the Safety Instrumented System (SIS), making it the proven pivot point of the TRITON/TRISIS pattern.

Architecture grouping (organized by ROLE within this layer, not by vendor)

Class (role)	Products (cross-vendor)	Stack (RTOS/protocol/OS)	Catalog depth
Chassis PLC (large process)	Rockwell ControlLogix/GuardLogix 1756-L7x/L8x + 1756-EN2T/EN3TR/EN4TR; Siemens SIMATIC S7-400, S7-1500 (CPU 151x); Schneider Modicon Quantum 140 / Premium TSX	VxWorks / proprietary; EtherNet/IP CIP; S7Comm (TCP/102); Modbus/UMAS TCP/502	Deep CVE coverage
Rack/compact PLC (mid-tier)	Rockwell CompactLogix/Compact GuardLogix 5380/5480/5370; Siemens S7-1200; Schneider Modicon M340/M580; Honeywell C300 / C200E Series-C	VxWorks; ARM/PowerPC RTOS; CIP / PROFINET / Modbus / CDA over UDP	Deep CVE coverage
Entry-level / micro PLC	Rockwell Micro800 (820/830/850/870); Siemens S7-1200 Basic; Schneider M221/M241	Lightweight RTOS; Modbus TCP/502; CIP; proprietary	Moderate CVE coverage
Distributed I/O / remote I/O	Siemens ET200SP / ET200MP / ET200ECO; Rockwell 1734-AENTR POINT I/O; Schneider	PROFINET RT/IRT; EtherNet/IP CIP; Modbus	Research only (low

Class (role)	Products (cross-vendor)	Stack (RTOS/protocol/OS)	Catalog depth
	Modicon X80 drops		public CVE volume)
RTU / utility comms processor	SEL-3530 / SEL-3555 RTAC; Schneider SCADAPack; Siemens SIMATIC RTU3030C; Honeywell ControlEdge RTU	Linux-based (RTAC); DNP3 TCP/20000; IEC 60870-5-104; IEC 61850	Moderate CVE coverage
Protective relay / IED (substation)	SEL-300/351/387/421/451/487/700/751; Siemens SIPROTEC 4/5; Schneider Sepam / MiCOM; ABB REF/REL/RET	Embedded RTOS; IEC 61850 GOOSE + MMS (TCP/102); DNP3; proprietary engineering	Deep CVE coverage
Variable-frequency drive (VFD)	Rockwell PowerFlex 525/527/753/755/755T/6000T; Siemens SINAMICS G120/S120/V90; Schneider Altivar	Embedded RTOS; CIP-Motion; PROFINET; Modbus; embedded web servers	Moderate CVE coverage

Group 1 — Chassis PLC (large process)

Direct attack surface: EtherNet/IP CIP (TCP/44818, UDP/2222 implicit messaging), S7Comm (TCP/102), Modbus TCP/502, embedded HTTP/HTTPS engineering ports, SNMP, FTP/TFTP on legacy comms cards. Studio 5000 / TIA Portal / Control Expert online sessions are authenticated by weak or proprietary schemes.

Confirmed CVEs across vendors:

CVE	CVSS	Vendor	Product	Vector
CVE-2021-22681	10.0	Rockwell	Logix family (1756/1769)	Hardcoded Studio 5000 cryptographic key — auth bypass, see ICSA-21-056-03
CVE-2022-1161	7.7	Rockwell	ControlLogix/CompactLogix/GuardLogix 5380/5480/5580	Stealthy logic injection — online view differs from running bytecode
CVE-2023-3596	7.5	Rockwell	1756-EN2T/EN3TR/EN4TR	Crafted CIP message → comms-card DoS
CVE-2024-6242	7.7	Rockwell	1756-EN4TR	CIP "trusted-slot" bypass — write to controller from untrusted slot

CVE	CVSS	Vendor	Product	Vector
Siemens ProductCERT advisories	varied	Siemens	S7-1500 / S7-1500F / S7-400	Multiple firmware advisories (CPU 151x denial-of-service, web-server XSS, MMS) — see ProductCERT monthly cycle
CVE-2018-7841	9.8	Schneider	Modicon Quantum / Premium / M580	UMAS auth bypass — unauthenticated stop/start/logic-write
Stuxnet (S7-300/400 PROFIBUS payload)	n/a	Siemens	S7-315/417 + CP 342-5	DLL hijack of s7otbxdx.dll, OB35 rewrite → centrifuge over-speed

Top attack vector (MITRE ATT&CK ICS): [T0833 Modify Control Logic](#), often chained with [T0843 Program Download](#) and [T0832 Manipulation of View](#).

Group 2 — Rack/compact PLC (mid-tier)

Direct attack surface: Same CIP/S7Comm/Modbus/UMAS stacks as Group 1 but typically fewer security features (no 1756-EN4TR trusted-slot enforcement; smaller key store). Honeywell C300 uses Control Data Access (CDA) over UDP — flat, unauthenticated, broadcast-rich.

Confirmed CVEs across vendors:

CVE	CVSS	Vendor	Product	Vector
CVE-2023-2071	9.8	Rockwell	CompactLogix 5370	Unauthenticated CIP RCE — see ICSA-23-136-04
CVE-2021-22779	8.1	Schneider	Modicon M340/M580/Quantum/Premium	UMAS reservation-bypass — write logic without engineering session
Honeywell Experion / C300 ICEFALL family	varied	Honeywell	Experion PKS / C300	CDA protocol weaknesses disclosed in the Clarity/Forescout OT:ICEFALL set (see Honeywell SN and Forescout writeup)
Siemens ProductCERT	varied	Siemens	S7-1200	Multiple denial-of-service via crafted S7Comm/web-server requests

CVE	CVSS	Vendor	Product	Vector
S7-1200 advisories				
CVE-2022-1161	7.7	Rockwell	CompactLogix 5380	Stealthy logic injection (also lists this device)

Top attack vector (MITRE ATT&CK ICS): [T0836 Modify Parameter](#), enabled by [T0866 Exploitation of Remote Services](#).

Group 3 — Entry-level / micro PLC

Direct attack surface: Modbus TCP/502 unauthenticated by spec; embedded web servers; cleartext configuration; many devices ship reachable on plant-wide VLANs. Shodan/Censys exposure is the dominant initial-access path.

Confirmed CVEs across vendors: Coverage is uneven — micro PLCs are under-disclosed relative to chassis siblings.

CVE	CVSS	Vendor	Product	Vector
CVE-2018-7841	9.8	Schneider	M221 (UMAS implementation)	Auth-bypass over Modbus/UMAS
Rockwell Micro800 advisories	varied	Rockwell	Micro820/830/850/870	Multiple Connected Components Workbench (CCW) CVEs published via Trust Center
Siemens ProductCERT S7-1200 advisories	varied	Siemens	S7-1200 Basic	Web-server / S7Comm DoS series

Top attack vector (MITRE ATT&CK ICS): [T0883 Internet Accessible Device](#) → [T0855 Unauthorized Command Message](#), often combined with [T0812 Default Credentials](#) on embedded web servers.

Group 4 — Distributed I/O / remote I/O

Direct attack surface: PROFINET RT (Ethernet 0x8892) and IRT are unauthenticated by IEC 61784 spec; DCP (Discovery and Configuration Protocol) supports SetName/SetIP without authentication. EtherNet/IP I/O drops trust any CIP producer on the segment.

Confirmed CVEs across vendors: Public CVE volume is low — most attacks are *protocol-level abuse* (PROFINET DCP name/IP hijack, CIP I/O spoofing) rather than firmware CVEs. Siemens ET200SP

firmware has periodic ProductCERT advisories (denial-of-service via crafted PROFINET frames).

Research anchor: [Claroty Team82 research](#).

Top attack vector (MITRE ATT&CK ICS): [T0830 Adversary-in-the-Middle](#) — protocol-spoofed I/O updates that desynchronize the PLC's image of physical state.

Group 5 — RTU / utility comms processor

Direct attack surface: DNP3 TCP/20000 (authenticated only when DNP3 Secure Authentication v5 is enabled — rarely is), IEC 60870-5-104 (TCP/2404), IEC 61850 MMS (TCP/102), embedded Linux on RTAC family with SSH/HTTPS management.

Confirmed CVEs across vendors:

CVE	CVSS	Vendor	Product	Vector
SEL RTAC advisories	varied	SEL	SEL-3530 / SEL-3555 RTAC	Multiple RTAC firmware issues (web UI, auth) — see SEL Security Advisories index
SEL acSELerator advisories	varied	SEL	acSELerator family	Engineering-tool CVEs that feed RTAC config
Schneider SCADAPack advisories	varied	Schneider	SCADAPack 32/350/470/474	PSIRT-listed DoS + auth issues

Top attack vector (MITRE ATT&CK ICS): [T0855 Unauthorized Command Message](#) (DNP3 / IEC 104 / IEC 61850 control-direction frames).

Group 6 — Protective relay / IED (substation)

Direct attack surface: IEC 61850 GOOSE (multicast, Ethertype 0x88B8 — *no auth, no encryption by base spec*) and MMS (TCP/102); DNP3; proprietary engineering protocols (SEL Fast Message, SIPROTEC DIGSI). Industroyer/Industroyer2 are the canonical adversary tooling.

Confirmed CVEs across vendors:

CVE	CVSS	Vendor	Product	Vector
SEL security advisories index	varied	SEL	300/351/387/421/451/487/700/751	Multiple firmware advisories — engineering protocol + web UI

CVE	CVSS	Vendor	Product	Vector
Siemens SIPROTEC advisories	varied	Siemens	SIPROTEC 4/5	DIGSI engineering session and IEC 61850 stack CVEs
Schneider Sepam/MiCOM advisories	varied	Schneider	Sepam 20/40/80, MiCOM P-series	PSIRT-listed firmware/web CVEs
ABB relay advisories	varied	ABB	REF/REL/RET 615/620/630 series	PSIRT-listed IEC 61850 and web-server CVEs
Industroyer2 (Dragos)	n/a	multi	IEC 61850 MMS / IEC 104	Direct breaker-open commands, attributed to SANDWORM

Top attack vector (MITRE ATT&CK ICS): [T0831 Manipulation of Control](#) via GOOSE injection or MMS control-block writes.

Group 7 — Variable-frequency drive (VFD)

Direct attack surface: Embedded web servers (often unauthenticated or default-credentialed), Modbus TCP/502, CIP-Motion, PROFINET. Drives are L1 *actuators-with-firmware*: a tampered drive will under/over-speed a motor regardless of correct PLC commands.

Confirmed CVEs across vendors:

CVE	CVSS	Vendor	Product	Vector
CVE-2018-19282	7.5	Rockwell	PowerFlex 525	Embedded web server — unauthenticated information disclosure / DoS
Siemens SINAMICS advisories	varied	Siemens	SINAMICS V90 / G120 / S120	Web server XSS, auth-bypass series via ProductCERT
Schneider Altivar advisories	varied	Schneider	Altivar Process / Machine	PSIRT-listed Modbus/web CVEs

Top attack vector (MITRE ATT&CK ICS): [T0879 Damage to Property](#) via [T0836 Modify Parameter](#) on drive setpoints.

Logging matrix (highest priority for this layer)

#	Source	Event	What it catches	MITRE ATT&CK ICS
1	OT IDS (Nozomi / Dragos / Claroty / Forescout SilentDefense)	CIP <code>Service 0x4B</code> (Forward_Open) to controller object, <code>Class 0x6B/0x68</code> (program download), S7Comm function <code>0x1A/0x1B/0x28/0x29</code> (PLC stop/start, block download)	Logic download or run-mode change	T0843 / T0858
2	OT IDS	Modbus function codes 8 (diagnostics), 43 (encapsulated UMAS), 90 (Schneider proprietary UMAS) from non-engineering host	Schneider UMAS auth-bypass attempts	T0855
3	PLC native audit log	Rockwell Logix Audit Source change-of-state; Siemens S7-1500 Security Event Log; Schneider M580 syslog	Mode key turned to REM, online edits, firmware update	T0858 / T0857
4	Engineering workstation (L3.5) EDR	Studio 5000 / TIA Portal / Control Expert / DIGSI / acSElerator project file open from unusual path or after email/download	Evil PLC reverse-attack pattern	T0817 / T0890
5	Switch / TAP	New MAC at PROFINET / EtherNet-IP segment; DCP SetName/SetIP frames; GOOSE traffic with new stNum jumps or duplicate goCBRef	Rogue device on L1 segment; GOOSE spoofing	T0830
6	Historian / SCADA	Setpoint write from non-operator workstation; setpoint write outside engineering window	Unauthorized command message to L1	T0855 / T0831
7	Network flow	New TCP/102, TCP/44818, TCP/502, TCP/20000, TCP/2404 talker; outbound from a PLC to anywhere	PLC-as-pivot (Evil PLC backchannel)	T0884 / T0822
8	Process historian	View-vs-reality delta: PLC reports nominal but downstream tag (flow, level, current) diverges	CVE-2022-1161 / Stuxnet view-vs-bytecode mismatch	T0832

Secondary:

- Vendor-native audit: Rockwell FactoryTalk AssetCentre disaster-recovery diff; Siemens TIA Portal Security Audit; Schneider EcoStruxure Cybersecurity Admin Expert (CAE) event log; SEL acSElerator audit trail.
- OT-native anomalies: PROFINET cycle-time jitter, CIP connection-count baseline drift, GOOSE stNum/sqNum baseline drift, drive parameter checksum baseline, IEC 61850 control-block (CTLNUM, OperOK) anomalies.

- Hardware: PLC keyswitch position telemetry (REM/PROG/RUN), chassis temperature/redundancy bit flips that often accompany firmware updates.

Cross-layer pivots

1. **L3 EWS → L1 PLC (Stuxnet pattern)**. A compromised engineering workstation at L3.5/L3 (Studio 5000, TIA Portal, Control Expert, DIGSI, acSElerator) issues authenticated logic downloads or parameter writes to the L1 controller. The session is indistinguishable from legitimate engineering. Detection lives on the EWS host (Logging matrix #4) and in CIP/S7Comm/UMAS payload inspection (#1, #2).
2. **L1 PLC → L0 actuator (the whole point)**. A logic-modified PLC writes manipulated setpoints/outputs to drives, breakers, valves — directly moving physical state. PowerFlex/SINAMICS/Altivar drive over-speed and SIPROTEC/SEL/Sepam breaker trip are the canonical destructive results ([T0879](#)).
3. **L1 → L3 Evil PLC reverse-attack (Claroty Team82)**. Attacker weaponizes the PLC project blob so that the next engineer to *open* a session is exploited and the engineering workstation is compromised — completing the loop in the opposite direction from Stuxnet. Applies to Rockwell, Siemens, Schneider engineering suites.
4. **L1 → L1 lateral via PROFINET / EtherNet-IP**. A compromised PLC abuses shared layer-2 to spoof PROFINET DCP, hijack ET200/POINT-I/O drops, or push CIP Forward_Open sessions to neighboring controllers and drives. The flat segment IS the trust boundary.
5. **L1 BPCS → SIS (TRITON pattern)**. Where the Basic Process Control System shares network with the Safety Instrumented System (Triconex Tricon, ProSafe-RS, AADvance, ControlEdge SC), a compromised L1 controller becomes the staging point to push firmware/logic into the SIS — the XENOTIME 2017 Saudi petrochemical playbook (Triconex TriStation protocol abuse).

Sources

- [CISA ICSA-21-056-03 — Rockwell Logix CVE-2021-22681](#)
- [CISA ICSA-23-136-04 — Rockwell CompactLogix 5370 CVE-2023-2071](#)
- [CISA AA24-038A — Volt Typhoon pre-positioning](#)
- [NVD CVE-2022-1161 — Rockwell stealthy logic injection](#)
- [NVD CVE-2018-7841 — Schneider Modicon UMAS](#)
- [NVD CVE-2021-22779 — Schneider Modicon UMAS reservation bypass](#)
- [NVD CVE-2023-3596 — Rockwell 1756-EN2T/EN3TR/EN4TR](#)
- [NVD CVE-2024-6242 — Rockwell 1756-EN4TR trusted-slot](#)
- [NVD CVE-2018-19282 — Rockwell PowerFlex 525](#)
- [NVD CVE-2023-2071 — Rockwell CompactLogix 5370](#)
- [Claroty Team82 research index](#)
- [Claroty Evil PLC Attack whitepaper](#)

- [Forescout OT:ICEFALL](#)
- [Dragos Industroyer2 analysis](#)
- [Dragos TRISIS / TRITON / XENOTIME analysis](#)
- [Rockwell Trust Center — security advisories](#)
- [Siemens ProductCERT](#)
- [Schneider Electric PSIRT — cybersecurity notifications](#)
- [Honeywell Cybersecurity Resources](#)
- [SEL Security Advisories](#)
- [ABB Cyber Security Alerts and Notifications](#)
- [MITRE ATT&CK for ICS — Techniques](#)
- [ISA/IEC 62443 series overview](#)
- [Purdue Enterprise Reference Architecture \(ISA-95\)](#)